

Skills Summary

Algebraic Limits: Factoring and Rationalizing

Use this reference while completing your worksheet or when studying.

Skill 1 — Limits by factoring (limits-by-factoring)

Rule:

If direct substitution gives $\frac{0}{0}$, the numerator and denominator share a factor of $(x - a)$. Factor, cancel it, then substitute.

$$a^2 - b^2 = (a - b)(a + b) \quad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Example: $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

Step 1. Substitution gives $\frac{0}{0}$, so factor rather than conclude.

Step 2. $\frac{(x - 4)(x + 4)}{x - 4} = x + 4$ for $x \neq 4$; at $x = 4$ this is 8.

Try it: $\lim_{x \rightarrow 5} \frac{x^2 - 2x - 15}{x - 5}$

check: 8

Skill 2 — Limits by rationalizing (limits-by-rationalizing)

Rule:

When a square root causes the $\frac{0}{0}$, multiply top and bottom by the **conjugate**. $(\sqrt{u} - v)(\sqrt{u} + v) = u - v^2$ clears the radical from the numerator and produces the cancelling factor.

Example: $\lim_{x \rightarrow 0} \frac{\sqrt{x + 1} - 1}{x}$

Step 1. Multiply by $\frac{\sqrt{x + 1} + 1}{\sqrt{x + 1} + 1}$ so the numerator becomes $(x + 1) - 1 = x$.

Step 2. Cancel x to get $\frac{1}{\sqrt{x + 1} + 1}$; at $x = 0$ this is $1/2$.

Try it: $\lim_{x \rightarrow 0} \frac{\sqrt{x + 9} - 3}{x}$

check: 1/9

Skill 3 — Removable discontinuities (removable-discontinuity)

Rule:

A piecewise f is continuous at $x = a$ exactly when $\lim_{x \rightarrow a} f(x) = f(a)$. To patch a hole, set the assigned value equal to the limit you computed.

Example: $f(x) = \frac{x^2 - 100}{x - 10}$ for $x \neq 10$, $f(10) = c$. Find c .

Step 1. Cancel: $\frac{(x - 10)(x + 10)}{x - 10} = x + 10$, so the limit at 10 is 20.

Step 2. Continuity forces $f(10)$ to equal that limit, so $c = \mathbf{20}$.

Try it: $f(x) = \frac{x^2 - 25}{x - 5}$ for $x \neq 5$, $f(5) = c$.

check: 10

(!) Watch out: Cancelling changes the function at the removed point, not the limit. Never write $\frac{0}{0} = 1$, and never conclude a limit fails to exist just because substitution is undefined.